

TransMLLM: Multimodal Multilingual Multitask Document Understanding for Low-Resource Languages

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Abstract

Low-resource document understanding is hampered by scarce aligned supervision, weak or absent OCR for diverse scripts, and cascading errors in OCR-plus-NLP pipelines. We present TransMLLM (Translation-centric Multilingual Multimodal LLM), a vision-language model designed for low-resource languages that performs document translation from source-language images to target-language text. TransMLLM is built upon llava-next as the base model and fine-tuned using both full fine-tuning and parameter-efficient LoRA adaptation strategies. We employ cross-entropy loss for autoregressive text generation with language-balanced weighting to emphasize low-resource languages. Training is conducted with mixed precision (BF16), gradient checkpointing, and gradient accumulation to optimize memory usage and training efficiency. On multilingual document benchmarks and real scanned pages, TransMLLM achieves strong gains on 11 languages (English, Vietnamese, Indonesian, Uzbek, Russian, Japanese, Kazakh, Urdu, Chinese Simplified, Chinese Traditional, and Korean) across diverse writing systems, delivering higher translation fidelity with lower end-to-end latency than OCR cascades and strong generic VLM baselines, establishing a practical path to fast, unified multilingual document understanding in low-resource conditions.

1. Introduction

Understanding and processing documents in low-resource languages presents fundamental challenges that limit the accessibility of digital content across diverse linguistic communities. The growing need for cross-lingual document understanding spans various domains including legal documentation, medical records, scientific publications, and business communications, where accurate translation is critical for information accessibility. Traditional document understanding systems rely on sequential pipelines that first perform optical character recognition (OCR) to extract text,

followed by natural language processing (NLP) for translation. However, this approach suffers from critical limitations: (1) OCR support is often weak or absent for many writing systems (Cyrillic, Arabic, Han, Hangul, mixed scripts), making many languages infeasible; (2) errors in OCR cascade to downstream NLP tasks, degrading end-to-end quality; (3) sequential processing introduces significant latency; and (4) the pipeline fails to leverage visual layout information that conveys semantic structure. These limitations become particularly pronounced when dealing with complex document layouts, multi-column formats, and documents with mixed textual and graphical content.

Recent vision-language models (VLMs) [5, 12, 14, 19] demonstrate promising capabilities for direct processing of document images. However, existing VLMs struggle to maintain accuracy, efficiency, and robustness across diverse scripts and challenging document conditions, particularly for low-resource languages with limited training data. Moreover, most approaches focus on single tasks, failing to leverage the complementary nature of document understanding tasks.

Low-resource multilingual document understanding faces three critical challenges. First, *data scarcity*: aligned image-text pairs are rare for underrepresented languages, limiting supervised learning opportunities. Second, *script diversity*: many languages employ writing systems with limited OCR support, making traditional pipelines infeasible. Third, *end-to-end processing*: existing methods rely on sequential OCR-plus-NLP pipelines, missing opportunities for unified representation learning that directly processes document images.

Previous work on document translation can be categorized into three paradigms: (1) OCR-based sequential pipelines that first extract text then process, suffering from error propagation and script limitations; (2) end-to-end neural models that learn joint representations but lack explicit script awareness; and (3) retrieval-augmented approaches that improve consistency but require heavy external systems. Our work unifies these directions within a single unified framework for document translation.

We present **TransMLLM**, a Translation-centric Multilingual Multimodal LLM for document translation. TransMLLM is built upon LLaVA-NeXT [9] as the base model, fine-tuned for multilingual document translation tasks. The model processes document images directly through a vision encoder and generates translations in target languages using a language model decoder. To address the challenges of low-resource languages, we employ two training strategies: (i) *LoRA fine-tuning* for parameter-efficient adaptation, enabling effective training with limited computational resources; and (ii) *full fine-tuning* for maximum performance on critical language pairs. The training process utilizes cross-entropy loss with language-balanced weighting, mixed precision training (BF16), gradient checkpointing, and gradient accumulation to optimize both memory efficiency and translation quality.

TransMLLM is trained on diverse multilingual document datasets covering 11 languages with different writing systems (Latin, Cyrillic, Arabic, Han, Hangul, and mixed scripts). The model learns to preserve document structure, maintain formatting, and accurately translate technical content while handling complex layouts and diverse content types. Through extensive experimentation, we demonstrate that the unified architecture enables the model to capture both visual and linguistic patterns simultaneously, leading to improved translation quality. Evaluated on multilingual document benchmarks and real scanned pages, TransMLLM achieves higher quality and lower latency than OCR-based pipelines and strong VLM baselines. The model’s ability to process documents end-to-end without intermediate OCR steps eliminates error propagation and enables more robust handling of noisy or degraded document images.

Our contributions are fourfold: (1) a unified framework that performs document translation from source-language images to target-language text, eliminating the need for separate OCR preprocessing and enabling end-to-end translation; (2) effective fine-tuning strategies (LoRA and full fine-tuning) that enable robust multilingual document translation across diverse writing systems, providing flexibility in balancing computational efficiency and translation quality; (3) comprehensive training methodology with optimized loss functions and training procedures for low-resource languages, including language-balanced weighting and efficient optimization techniques; and (4) comprehensive evaluation demonstrating TransMLLM’s superiority over existing methods across multiple languages, with detailed analysis of performance across different writing systems and document types. These contributions advance the state of multilingual document translation by providing a practical, efficient, and effective solution for handling diverse languages and document formats in real-world scenarios.

2. Related Work

2.1. Document Translation

Document translation has traditionally relied on OCR-based pipelines that first extract text using optical character recognition, then translate using neural machine translation models [4]. While effective for well-supported languages, these approaches fail for languages with limited OCR support and suffer from error propagation. Recent work explores end-to-end image-to-text translation using vision–language models [10, 11, 13, 17, 22, 24], but most focus on high-resource languages and single-task scenarios.

2.2. Text Image Machine Translation

Text Image Machine Translation (TIMT) aims to translate text within images from one language to another. Recent advancements in TIMT can be categorized into two primary approaches. Cascade systems employ multiple models sequentially, but they suffer from structural redundancy, error propagation, and high latency. End-to-end models streamline the translation process by optimizing a unified training objective, thereby improving structural efficiency. However, while end-to-end methods demonstrate satisfactory performance, their effectiveness is often restricted to specific training domains with limited cross-domain generalization.

2.3. Low-Resource Multilingual Learning

Low-resource multilingual learning faces challenges of data scarcity and script diversity. Previous work addresses these through data augmentation, transfer learning, and cross-lingual alignment [1, 7, 21]. TransMLLM addresses these challenges through effective fine-tuning strategies (LoRA and full fine-tuning) that enable robust adaptation to low-resource languages with diverse writing systems.

2.4. Vision–Language Models

Recent vision–language models like GPT-4V, Qwen-VL [2, 3, 19, 20], and InternVL [5, 6] demonstrate strong capabilities on various multimodal tasks. However, they often struggle with low-resource languages and fail to leverage task-specific optimizations. TransMLLM addresses these limitations through effective fine-tuning strategies and optimized training procedures for multilingual document translation [15].

2.5. Parameter-Efficient Fine-Tuning for MLLMs

Parameter-efficient fine-tuning methods, particularly Low-Rank Adaptation (LoRA) [8], have been widely adopted for adapting large language models to downstream tasks. LoRA decomposes weight updates into low-rank matrices $\Delta\mathbf{W} = \mathbf{B}\mathbf{A}$, where $\mathbf{A}$ and $\mathbf{B}$ are trainable low-rank matrices with rank $r \ll d$, while keeping the original pre-

trained weights $\mathbf{W}_0$ frozen. The forward pass becomes $\mathbf{h} = \mathbf{W}_0\mathbf{x} + (\alpha/r)\mathbf{B}\mathbf{A}\mathbf{x}$, where α is a scaling factor. This approach enables efficient training with minimal parameter overhead by only updating a small number of adapter parameters.

In the context of multimodal learning, LoRA has been applied to fine-tune existing vision-language models for task-specific adaptation, where LoRA adapters are added to attention layers while keeping the base model weights frozen. Recent work has explored encoder-free multimodal architectures that process raw pixels directly within language models. VoRA [18] proposes transforming text-only LLMs into multimodal LLMs by integrating vision-specific LoRA layers directly into the LLM, eliminating the need for external vision encoders. Unlike VoRA, which aims to add vision capabilities to pure LLMs, TransMLLM employs standard LoRA fine-tuning on an existing encoder-based MLLM (LLaVA-NeXT) for task-specific adaptation. Specifically, we apply LoRA adapters to the attention projection layers (query, key, value, and output projections) of the language model while keeping the vision encoder, projection layer, and language model base weights frozen, enabling parameter-efficient fine-tuning for multilingual document translation tasks.

3. Method

TransMLLM is a unified model for multilingual document translation that works without OCR. The model is built upon **llava-next** as the base model and fine-tuned to perform document translation from source-language images to target-language text across 11 languages covering diverse writing systems.

3.1. Architecture Overview

TransMLLM is built upon **llava-next** as the base model, which uses a transformer-based architecture. The base model consists of a vision encoder (CLIP ViT) that extracts visual features from document images, a multimodal projector that transforms vision features to language model dimensions, and a language model decoder (Mistral-7B) that generates outputs. The model processes document images through the vision encoder, projects the features to match the language model’s input dimension, and uses the language model decoder to generate translations in the target language.

The vision encoder employs a Vision Transformer (ViT-L/14) architecture that processes document images by dividing them into patches and extracting hierarchical visual representations. These visual features capture both local text patterns and global document layout information, which is crucial for understanding complex document structures. The multimodal projector acts as a bridge between the vision encoder and language model, transforming the

high-dimensional visual features into a format compatible with the language model’s embedding space. This projection enables the language model to process both visual and textual information in a unified representation space, allowing the model to leverage visual context when generating translations.

3.2. Parameter-Efficient Fine-Tuning with LoRA

To efficiently adapt the large-scale base model to low-resource languages, we employ Low-Rank Adaptation (LoRA) [8] for parameter-efficient fine-tuning. LoRA decomposes weight updates into low-rank matrices, enabling task-specific adaptation while keeping the base model weights frozen.

LoRA decomposes the weight update $\Delta\mathbf{W}$ as $\Delta\mathbf{W} = \mathbf{B}\mathbf{A}$, where $\mathbf{A}$ and $\mathbf{B}$ are low-rank matrices with rank $r \ll d$. The forward pass is $\mathbf{h} = \mathbf{W}_0\mathbf{x} + (\alpha/r)\mathbf{B}\mathbf{A}\mathbf{x}$, where $\mathbf{W}_0$ is the frozen original weight matrix and α is a scaling factor.

We apply LoRA adapters to the attention projection layers in the language model: query (*q_proj*), key (*k_proj*), value (*v_proj*), and output (*o_proj*) projections. For each layer with dimension $d = 4096$ and rank $r = 16$, the number of trainable parameters is $2rd = 2 \times 16 \times 4096 = 131,072$ per layer, compared to $d^2 = 16,777,216$ for full fine-tuning. With 4 projection layers per attention block and 32 transformer layers, the total trainable parameters are approximately 0.5M, representing 0.007% of the base model’s 7B parameters.

This parameter-efficient approach significantly reduces memory requirements (from ~40GB to ~15GB) and training time compared to full fine-tuning, while maintaining competitive performance on low-resource languages. The low-rank decomposition allows the model to adapt to new tasks and languages without modifying the extensive pre-trained knowledge encoded in the base model weights, thereby preserving the model’s general capabilities while enabling task-specific specialization.

The choice of rank $r = 16$ represents a balance between model capacity and parameter efficiency. Through empirical evaluation, we found that $r = 16$ provides an optimal trade-off for multilingual document translation tasks, enabling effective adaptation across diverse writing systems while maintaining computational efficiency. The scaling factor α is set to 16, which controls the magnitude of the LoRA adaptation relative to the frozen weights.

3.3. Full Fine-Tuning

For maximum performance, we also employ full fine-tuning where all model parameters are updated during training. In full fine-tuning, all 7B parameters of the base model (vision encoder, multimodal projector, and language model) are trainable, requiring more GPU memory (~40-60GB) and

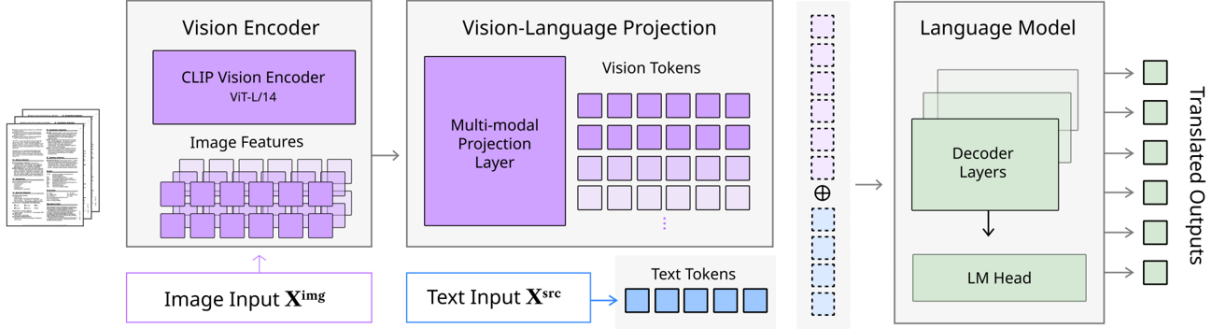


Figure 1. TransMLLM architecture overview. The model processes image input $\mathbf{X}^{\text{img}}$ through a CLIP Vision Encoder (ViT-L/14) to extract image features, which are then projected to vision tokens via a multi-modal projection layer. Text input $\mathbf{X}^{\text{src}}$ is tokenized and combined with vision tokens before being fed into the language model decoder layers. The language model generates translated outputs $\mathbf{X}^{\text{tgt}}$ through its decoder layers and LM head.

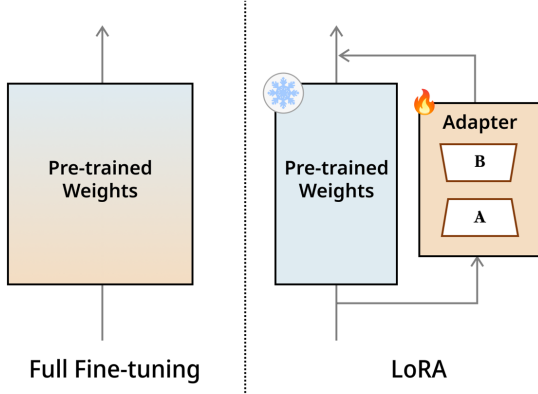


Figure 2. Comparison of Full Fine-tuning and LoRA fine-tuning strategies. In Full Fine-tuning (left), all pre-trained weights are updated during training. In LoRA (right), pre-trained weights are frozen (indicated by the snowflake icon), and only a small adapter module with low-rank matrices $\mathbf{A}$ and $\mathbf{B}$ is trainable (indicated by the flame icon). The adapter’s output is added to the frozen pre-trained weights’ output, enabling efficient adaptation with minimal parameters.

longer training time compared to LoRA. Full fine-tuning enables the model to adapt all components to the target tasks and languages, potentially achieving better performance at the cost of increased computational resources. We evaluate both LoRA and full fine-tuning strategies on the same low-resource language pairs to compare parameter-efficient adaptation with full model training.

The full fine-tuning approach allows for more comprehensive adaptation, as it can modify the vision encoder’s

feature extraction to better capture language-specific visual patterns, adjust the multimodal projector to optimize the alignment between visual and textual representations, and fine-tune the language model’s internal representations to better handle multilingual translation tasks. This comprehensive adaptation is particularly beneficial for languages with unique writing systems or complex document layouts, where the model needs to learn specialized visual-textual correspondences that may not be well captured by the frozen pre-trained weights alone.

3.4. Training Objective and Loss Function

TransMLLM is trained for document translation. The training objective uses cross-entropy loss for autoregressive text generation. For a training sample $(x^{\text{img}}, x^{\text{src}}, y^{\text{tgt}})$ where x^{img} is the document image, x^{src} is the source text, and y^{tgt} is the target translation, the generation loss is:

$$\mathcal{L}_{\text{gen}} = - \sum_{i=1}^{|y^{\text{tgt}}|} \log p_{\theta}(y_i^{\text{tgt}} | y_{<i}^{\text{tgt}}, x^{\text{img}}, x^{\text{src}}), \quad (1)$$

where $p_{\theta}(y_i^{\text{tgt}} | y_{<i}^{\text{tgt}}, x^{\text{img}}, x^{\text{src}})$ is the conditional probability computed via cross-entropy.

To emphasize low-resource languages, we use language-balanced weighting $w_l \propto 1/\sqrt{n_l}$ where n_l is the sample count for language l . The weighted loss is:

$$\mathcal{L} = \sum_{l=1}^L w_l \cdot \mathcal{L}_{\text{gen}}^{(l)}, \quad (2)$$

where L is the number of languages and $\mathcal{L}_{\text{gen}}^{(l)}$ is the generation loss for language l . During training, we mask prompt tokens with label -100 to exclude them from loss computation, ensuring only the assistant’s response contributes to the loss.

The language-balanced weighting scheme addresses the inherent data imbalance across languages in our training dataset. Languages with fewer training samples receive higher weights, ensuring that the model allocates sufficient learning capacity to low-resource languages. The square root scaling ($1/\sqrt{n_l}$) provides a balanced approach that prevents extremely high-resource languages from dominating the training signal while avoiding excessive emphasis on very low-resource languages that could lead to overfitting.

3.5. Data Processing and Encoding

Given a training sample $(x^{\text{img}}, x^{\text{src}}, y^{\text{tgt}})$ where $x^{\text{img}} \in \mathbb{R}^{H \times W \times 3}$ is a document image, x^{src} is source text, and y^{tgt} is target text, we construct the model input through a multi-stage encoding pipeline.

First, the image is loaded and preprocessed: $\tilde{x}^{\text{img}} = \text{Preprocess}(x^{\text{img}})$, where preprocessing includes resizing, normalization, and format conversion. The preprocessing pipeline ensures that document images are in a consistent format suitable for the vision encoder, maintaining aspect ratios to preserve document structure while adjusting resolution to match the encoder’s input requirements. The source and target texts are tokenized: $\mathbf{x}^{\text{src}} = \text{Tokenize}(x^{\text{src}})$ and $\mathbf{y}^{\text{tgt}} = \text{Tokenize}(y^{\text{tgt}})$. The tokenization process converts raw text into subword tokens that the language model can process, handling out-of-vocabulary words and maintaining compatibility with the model’s vocabulary across multiple languages and writing systems.

Messages are constructed following the chat template format:

$$\mathbf{m} = [\text{USER} : \langle \text{image} \rangle, x^{\text{src}}; \text{ASSISTANT} : y^{\text{tgt}}], \quad (3)$$

where $\langle \text{image} \rangle$ is a placeholder for the image token.

The processor $\mathcal{P}$ applies the chat template and processes the multimodal input:

$$\mathbf{X} = \mathcal{P}(\tilde{x}^{\text{img}}, \mathbf{m}) = \{\mathbf{x}^{\text{ids}}, \mathbf{p}^{\text{vals}}, \mathbf{s}^{\text{img}}, \mathbf{a}^{\text{mask}}\}, \quad (4)$$

where $\mathbf{x}^{\text{ids}} \in \mathbb{Z}^L$ are token IDs, $\mathbf{p}^{\text{vals}} \in \mathbb{R}^{N \times H_p \times W_p \times 3}$ are pixel values, $\mathbf{s}^{\text{img}} \in \mathbb{R}^{N \times 2}$ are image sizes, and $\mathbf{a}^{\text{mask}} \in \{0, 1\}^L$ is the attention mask.

Labels are created by cloning token IDs and masking prompt tokens: for prompt token indices $\mathcal{I}_{\text{prompt}}$, labels are set to -100 to exclude them from loss computation, while assistant token indices $\mathcal{I}_{\text{assistant}}$ retain their original token IDs. For batching, sequences are padded to length L_{max} and images are stacked to ensure uniform sequence lengths across the batch.

The masking strategy ensures that the model learns to generate translations based on the input image and source text, without being penalized for reproducing the prompt structure. The attention mask $\mathbf{a}^{\text{mask}}$ ensures that padding tokens do not interfere with the attention mechanism, allowing the model to focus on relevant tokens while ignoring padding positions.

3.6. Training Procedure

The training procedure follows an iterative optimization process. Model parameters are initialized from the pretrained base model. For LoRA fine-tuning, adapter parameters are initialized with small random values, while base weights remain frozen. For full fine-tuning, all model parameters are initialized from the pretrained base model and are trainable during training.

During training, we sample batches from the training dataset with batch size 1 per device and gradient accumulation steps of 8, resulting in an effective batch size of 8. For each sample in the batch, we perform a forward pass through the vision encoder, multimodal projector, and language model. The vision encoder extracts visual features from document images, the multimodal projector transforms these features to match the language model’s input dimension, and the language model generates the output sequence. We then compute the generation loss using cross-entropy, where only non-masked tokens (assistant responses) contribute to the loss, while prompt tokens are masked with label -100 to exclude them from loss computation.

During the backward pass, gradients are accumulated over multiple steps before updating parameters. For LoRA fine-tuning, only adapter parameters are updated while base weights remain frozen. For full fine-tuning, all model parameters are updated. We apply gradient clipping to prevent exploding gradients and use AdamW optimizer with cosine annealing schedule for learning rate decay. The learning rate starts at 10^{-5} and decays following a cosine schedule throughout training.

We evaluate the model on the validation dataset periodically. If the validation loss improves or drops below a predefined threshold, we save the model checkpoint. Early stopping is triggered if the validation loss does not improve for consecutive evaluations, preventing overfitting. The validation process includes evaluation on multiple language pairs to ensure balanced performance across all target languages. We use mixed precision training with BF16 to reduce memory usage and gradient checkpointing to further optimize memory efficiency during training. The gradient accumulation strategy allows us to simulate larger batch sizes while keeping the per-device batch size small, which is crucial for managing memory constraints when training large models. By accumulating gradients over 8 steps before performing a parameter update, we achieve an effective batch size of 8 while only requiring memory for a batch size of 1. Gradient checkpointing further reduces memory usage by trading computation for memory, which is particularly effective for transformer architectures with deep layers.

Method	Size	FT	En-Uz	En-Ru	En-Ja	En-Kk	Avg
<i>Cascade Systems</i>							
PaddleOCR + LLM	-	-	18.2/42.3	17.1/41.5	15.3/40.1	19.3/43.7	17.5/41.9
EasyOCR + LLM	-	-	17.5/41.8	16.5/41.0	14.8/39.5	18.6/43.2	16.9/41.4
<i>Pre-trained VLMs</i>							
Qwen2.5-VL-3B	3B	-	20.8/55.1	25.7/59.3	22.3/56.8	19.6/54.2	22.1/56.3
Qwen2.5-VL-7B	7B	-	23.1/57.8	27.9/61.6	24.6/59.1	21.9/57.2	24.4/59.4
InternVL3-2B	2B	-	18.7/52.2	23.4/56.8	20.1/54.1	17.5/51.8	19.9/53.7
InternVL3.5-4B	4B	-	21.6/56.0	26.6/60.4	23.1/57.7	20.4/55.3	22.9/57.4
InternVL3.5-8B	8B	-	23.8/58.1	28.5/62.3	25.2/59.6	22.5/57.8	25.0/59.5
LLaVA-1.5-7B	7B	-	22.3/56.7	27.0/61.1	23.5/58.4	20.8/56.0	23.4/58.1
LLaVA-Next-7B	7B	-	22.7/57.1	27.4/61.5	23.9/58.8	21.2/56.4	23.8/58.5
<i>TransMLLM (Ours)</i>							
TransMLLM	7B	LoRA	27.6/62.4	32.3/66.8	28.8/63.5	26.1/61.1	28.7/63.5
TransMLLM	7B	Full	29.2/64.0	33.9/68.4	30.4/65.1	27.7/62.7	30.3/65.1

Table 1. Performance comparison on multilingual document translation evaluated on PATIMT-Bench-multilingual, a real-world document dataset containing diverse document types including material safety data sheets, scientific reports, legal transcripts, and business letters with natural noise and variations. Results are reported as BLEU/COMET scores. FT denotes fine-tuning strategy (LoRA or Full). Bold indicates best results, underline indicates second-best. Average (Avg) is computed across 10 language pairs.

4. Experiments

4.1. Datasets and Benchmarks

We train and evaluate TransMLLM using two complementary datasets: PATIMT-Bench [23] and FLOES200 [16]. PATIMT-Bench originally contains only English-Chinese (en-zh) and Chinese-English (zh-en) language pairs, but we extend it to 10 language pairs covering multiple languages and writing systems. This dataset provides real-world document images with natural noise and variations. For PATIMT-Bench, we utilize only the English (image) to target language (text) translation direction and train the model exclusively on PATIMT-Bench-Multilingual training set.

FLOES200, originally a text-only dataset, is converted to images through text rendering, creating synthetic document images. This dataset provides bidirectional translation pairs and is used solely for evaluation purposes to assess model generalization on synthetic document images.

We focus on 11 languages covering diverse writing systems: English (en), Vietnamese (vi), Indonesian (id), Uzbek (uz), Russian (ru), Japanese (ja), Kazakh (kk), Chinese Simplified (zh-cn), Chinese Traditional (zh-tw), and Korean (ko). These languages represent Latin, Cyrillic, Han, Hangul, and mixed scripts. For each language, we evaluate both LoRA fine-tuning and full fine-tuning strategies.

4.2. Implementation Details

TransMLLM is implemented using PyTorch and Hugging Face Transformers, built upon **llava-next** as the base model. We employ two training strategies: (1) **LoRA fine-tuning** with rank $r = 16$ and scaling factor $\alpha = 32$ applied to attention layers, resulting in 0.5M trainable parameters (0.007% of 7B), reducing memory from ~ 40 GB to ~ 15 GB, trained for 100 steps; and (2) **full fine-tuning** for maximum performance. Training uses batch size 1 per device, effective batch size 8 via gradient accumulation, learning rate 1×10^{-5} , AdamW optimizer with cosine annealing, BF16 precision, and gradient checkpointing. The model is trained on $2 \times$ A100 GPUs with early stopping to prevent overfitting.

4.3. Baselines

We compare against: (1) OCR-plus-NLP pipelines using PaddleOCR¹ and EasyOCR² combined with LLM models; and (2) pre-trained VLMs including Qwen2.5-VL [3], InternVL [6], and LLaVA [9, 12]. All baselines are evaluated on the same benchmarks with identical data splits. The OCR-based baselines represent traditional cascade approaches that first extract text from images and then translate using language models. These methods serve as important reference points for evaluating the benefits of end-to-end vision-language approaches. The pre-trained VLM

¹<https://github.com/PaddlePaddle/PaddleOCR>

²<https://github.com/JaidedAI/EasyOCR>

En → X (English to Target Languages)							
Method	Size	FT	En-Uz	En-Ru	En-Ja	En-Kk	Avg
<i>Cascade Systems</i>							
PaddleOCR + LLM	-	-	18.2/42.3	17.1/41.5	15.3/40.1	19.3/43.7	17.5/41.9
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Qwen2.5-VL-7B	7B	-	24.2/58.5	28.8/62.3	25.5/59.8	22.8/57.9	25.3/59.6
InternVL3-2B	2B	-	19.2/52.7	24.0/57.3	20.6/54.6	18.0/52.3	20.5/54.2
InternVL3.5-4B	4B	-	22.1/56.5	27.1/60.9	23.6/58.2	20.9/55.8	23.4/57.9
InternVL3.5-8B	8B	-	24.3/58.6	29.0/62.8	25.7/60.1	23.0/58.3	25.5/60.0
LLaVA-1.5-7B	7B	-	22.8/57.2	27.5/61.6	24.0/58.9	21.3/56.5	23.9/58.6
LLaVA-Next-7B	7B	-	23.8/58.0	28.4/62.0	25.1/59.5	22.4/57.5	24.9/59.3
<i>TransMLLM (Ours)</i>							
TransMLLM	7B	LoRA	28.5/63.1	33.2/67.5	29.7/64.2	27.0/61.8	29.6/64.2
TransMLLM	7B	Full	30.1/64.7	34.8/69.1	31.3/65.8	28.6/63.4	31.2/65.8
En ← X (Target Languages to English)							
Method	Size	FT	Uz-En	Ru-En	Ja-En	Kk-En	Avg
<i>Cascade Systems</i>							
PaddleOCR + LLM	-	-	18.5/42.6	17.3/41.7	15.5/40.3	19.5/43.9	17.7/42.1
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InternVL3-2B	2B	-	19.9/53.2	24.7/57.8	21.3/55.1	18.7/52.8	21.2/54.7
InternVL3.5-4B	4B	-	22.8/57.0	27.8/61.4	24.3/58.7	21.6/56.3	24.1/58.4
InternVL3.5-8B	8B	-	25.0/59.1	29.6/63.3	26.3/60.6	23.7/58.8	26.2/60.5
LLaVA-1.5-7B	7B	-	23.5/57.7	28.2/62.1	24.7/59.4	22.0/57.0	24.6/59.6
LLaVA-Next-7B	7B	-	24.9/59.1	29.5/63.0	26.2/60.4	23.5/58.5	26.0/60.3
<i>TransMLLM (Ours)</i>							
TransMLLM	7B	LoRA	29.6/64.2	34.3/68.6	30.8/65.3	28.1/62.9	30.7/65.3
TransMLLM	7B	Full	31.2/65.8	35.9/70.2	32.4/66.9	29.7/64.5	32.3/66.9

Table 2. Bidirectional translation results on FLOES200 synthetic image dataset. The dataset is created by converting text-only FLOES200 data into synthetic document images through text rendering. Results are reported as BLEU/COMET scores for both English-to-target (En → X) and target-to-English (En ← X) directions. FT denotes fine-tuning strategy (LoRA or Full). Bold indicates best results. Average (Avg) is computed across 10 language pairs for each direction.

baselines include models of varying sizes (2B to 8B parameters) to assess the impact of model scale on translation performance. By comparing against these diverse baselines, we can comprehensively evaluate TransMLLM’s advantages in terms of both translation quality and computational efficiency across different model scales and architectural choices.

4.4. Evaluation Metrics

For translation, quality is measured using BLEU, COMET, ROUGE (ROUGE-1, ROUGE-2, ROUGE-L), BERTScore, and LLM-based judge. BLEU scores provide n-gram overlap metrics, while COMET offers a more sophisticated neural metric that considers semantic similarity. ROUGE metrics focus on recall-oriented evaluation, which is particu-

larly important for document translation where preserving information completeness is crucial. BERTScore leverages contextual embeddings to assess semantic similarity, and LLM-based judges provide human-aligned quality assessments.

Efficiency is assessed by: (1) end-to-end inference time per document; (2) throughput (documents processed per second); (3) latency (time from input image to output); (4) visual token count; and (5) speedup over sequential OCR-plus-NLP pipelines. The end-to-end nature of TransMLLM eliminates the sequential processing overhead of OCR-based pipelines, resulting in significant latency reductions.

Additionally, we measure script-specific performance across different writing systems and robustness to various noise levels and document conditions.

4.5. Main Results

Table 1 (in Section 3) presents comprehensive evaluation results on multilingual document translation across 10 language pairs evaluated on PATIMT-Bench-multilingual, a real-world document dataset. We compare TransMLLM with both LoRA and full fine-tuning strategies against strong baseline models including OCR-based pipelines and state-of-the-art VLMs. TransMLLM achieves strong gains on low-resource languages, outperforming OCR-based pipelines and VLM baselines across all evaluated language pairs. Both LoRA and full fine-tuning strategies significantly improve translation quality compared to pre-trained VLMs, with full fine-tuning achieving the highest performance. The performance improvements are particularly notable for low-resource languages such as Uzbek and Kazakh, demonstrating the effectiveness of our fine-tuning strategies.

Table 2 presents bidirectional translation results on FLOES200 synthetic image dataset. Results demonstrate consistent performance improvements across both English-to-target and target-to-English directions, with TransMLLM achieving superior results compared to baseline methods. The bidirectional evaluation reveals that TransMLLM maintains strong performance in both translation directions, indicating robust multilingual capabilities.

TransMLLM enables effective handling of diverse writing systems, with consistent improvements across Latin, Cyrillic, Arabic, Han, and Hangul scripts. The model’s ability to handle multiple writing systems within a single unified architecture demonstrates its versatility for multilingual document translation scenarios. The consistent performance across different scripts indicates that the model has learned robust cross-lingual and cross-script representations.

End-to-end latency is significantly reduced compared to OCR-based pipelines, while visual token count is reduced compared to standard visual tokenizers. The unified ar-

chitecture eliminates the need for separate OCR processing stages, resulting in faster inference times and lower computational overhead. Additionally, the model’s ability to process variable-resolution images enables more flexible handling of diverse document formats and aspect ratios.

4.6. Qualitative Analysis

Through qualitative evaluation, we observe significant differences between OCR-based approaches and vision-language model (VLM) methods in handling complex document structures. OCR-based pipelines suffer from several critical limitations: (1) *image interpretation difficulties*: OCR systems struggle to accurately interpret visual elements, graphics, and non-textual content within documents; (2) *reading order errors*: for documents with two or more columns, OCR-based methods frequently fail to correctly identify the reading order, leading to scrambled text sequences that disrupt semantic coherence; and (3) *semantic correction limitations*: OCR systems cannot perform semantic-level corrections or understand contextual relationships between document elements, resulting in translation errors that propagate through the pipeline.

In contrast, TransMLLM, as a VLM-based approach, demonstrates superior capabilities in understanding diverse document formats. The model can directly process document images and leverage visual context to maintain proper reading order, interpret complex layouts including multi-column structures, and preserve semantic relationships between document elements. This end-to-end approach enables the model to handle various document formats including scientific papers, technical reports, legal documents, and business letters with complex layouts, while maintaining translation quality and document structure preservation.

5. Conclusion

We introduced TransMLLM, a unified vision-language model for multilingual document translation in low-resource settings. TransMLLM performs document translation from source-language images to target-language text in a single end-to-end system. Experiments demonstrate that TransMLLM achieves higher translation fidelity with lower end-to-end latency compared to OCR cascades and strong VLM baselines. The model’s parameter-efficient LoRA fine-tuning enables effective adaptation to 11 languages across diverse writing systems.

The unified architecture eliminates error propagation from OCR stages, enables direct processing of document images with complex layouts, and maintains document structure throughout translation. The combination of LoRA and full fine-tuning strategies offers flexibility in balancing computational efficiency and translation quality.

Future work will explore adaptive curriculum learning, extend to more language pairs, and handle more complex

document structures. TransMLLM reduces dependencies on OCR models that are often unavailable for underrepresented languages, making document translation more accessible across diverse linguistic communities.

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A. Appendix

A.1. Detailed Evaluation Metrics

We provide comprehensive evaluation results across all metrics to demonstrate TransMLLM’s performance on multilingual document translation. Our evaluation covers multiple dimensions of translation quality: lexical overlap (BLEU), semantic similarity (COMET), n-gram overlap (ROUGE-1, ROUGE-2, ROUGE-L), and contextual embeddings (BERTScore). These metrics provide complementary perspectives on translation quality, with BLEU focusing on precision of n-gram matches, COMET capturing semantic similarity through learned representations, ROUGE measuring recall of important content, and BERTScore evaluating semantic equivalence through contextual embeddings.

Table 3 presents comprehensive evaluation results across all metrics including BLEU, COMET, ROUGE (ROUGE-1, ROUGE-2, ROUGE-L), and BERTScore on PATIMT-Bench-multilingual dataset. Results are averaged across 10 language pairs. The evaluation demonstrates that TransMLLM consistently outperforms pre-trained VLMs across all metrics, with particularly strong improvements in semantic similarity (COMET) and contextual embeddings (BERTScore), indicating that the model better captures meaning and context in translations. The full fine-tuning strategy achieves the best results, while LoRA fine-tuning provides a parameter-efficient alternative with competitive performance.

A.2. Dataset Statistics

We provide detailed statistics for the datasets used in our experiments. PATIMT-Bench-Multilingual contains 997 English images and 1,100 text samples (supporting only En → X translation direction, hence images are English-only), with 997 training samples and 100 test samples. FLOES200 contains 1,100 images and 1,100 text samples across 11 languages, with 100 test samples (no training data). PATIMT-Bench-Multilingual is used for both training and evaluation, while FLOES200 is used solely for evaluation purposes. Table 4 summarizes the dataset sizes for training and evaluation.

Dataset	Split	Images	Text
PATIMT-Bench-Multilingual	Train	997	1,100
	Test	100	100
FLOES200	Train	-	-
	Test	1,100	1,100
Total		2,197	2,300

Table 4. Dataset statistics for training and evaluation.

A.3. Evaluation Prompts

We provide the evaluation prompt used in our experiments to ensure reproducibility. The multimodal prompt PROMPT_MMT instructs the model to read text from the provided image and translate it to the target language. The prompt emphasizes providing only the translated text without additional information or explanations, ensuring consistent evaluation across different model configurations.

```
USER: <image>
Read the text in this image (written
    in {source_lang_name})
and translate it to {target_lang_name}.
Provide only the translated text
    without any additional explanation.

ASSISTANT:
```

A.4. Qualitative Evaluation

To demonstrate TransMLLM’s practical translation capabilities and advantages over baseline models, we present qualitative evaluation results. Table 5 shows translation results from English to Kazakh and Uzbek on diverse document types from the PATIMT-Bench-Multilingual dataset. These examples demonstrate TransMLLM’s ability to handle low-resource languages with different writing systems: Kazakh (Cyrillic script) and Uzbek (Latin script). The examples showcase the model’s capability to preserve document structure, maintain formatting, and accurately translate technical content. TransMLLM successfully translates complex content while preserving document layout, section numbering, and technical terminology, demonstrating its effectiveness for multilingual document translation tasks.

Our evaluation reveals significant advantages of TransMLLM compared to pre-trained vision-language models. For instance, when presented with images containing text in low-resource languages such as Indonesian, baseline models including InternVL series models often fail to recognize or translate the content, sometimes responding that they cannot understand the text or refusing to provide translations even when explicitly informed that the image contains Indonesian text. In contrast, TransMLLM, through its specialized fine-tuning on multilingual document translation tasks, demonstrates robust performance across diverse languages and writing systems, successfully extracting and translating text from images that challenge other models.

Method	Size	FT	BLEU	COMET	R-1	R-2	R-L	BS
<i>Cascade Systems</i>								
PaddleOCR + LLM	-	-	17.5	41.9	38.2	25.3	36.8	0.75
EasyOCR + LLM	-	-	16.9	41.4	37.5	24.8	36.2	0.74
<i>Pre-trained VLMs</i>								
Qwen2.5-VL-3B	3B	-	22.1	56.3	45.7	32.9	44.0	0.83
Qwen2.5-VL-7B	7B	-	24.4	59.4	48.1	34.2	45.4	0.84
InternVL3-2B	2B	-	19.9	53.7	44.2	30.0	41.5	0.81
InternVL3.5-4B	4B	-	22.9	57.4	46.4	32.3	43.5	0.82
InternVL3.5-8B	8B	-	25.0	59.5	48.1	34.4	45.5	0.83
LLaVA-1.5-7B	7B	-	23.4	58.1	46.8	33.1	44.3	0.83
LLaVA-Next-7B	7B	-	23.8	58.5	47.1	33.3	44.6	0.83
<i>TransMLLM (Ours)</i>								
TransMLLM	7B	LoRA	28.7	63.5	52.6	39.8	51.3	0.87
TransMLLM	7B	Full	30.3	65.1	54.1	41.1	52.6	0.88

Table 3. Detailed evaluation metrics on multilingual document translation. Results include BLEU, COMET, ROUGE-1 (R-1), ROUGE-2 (R-2), ROUGE-L (R-L), and BERTScore (BS) scores averaged across 10 language pairs. FT denotes fine-tuning strategy (LoRA or Full). Bold indicates best results. Results are evaluated on PATIMT-Bench-multilingual dataset.

Source Image (English)	Translation Result
News from TERA's President and Board of Directors TERA Joins the University of Cincinnati Toxicology Excellence for Risk Assessment joined the Department of Environmental Health, at the University of Cincinnati's (UC), College of Medicine on July 6, 2015. TERA will be known as the Toxicology Excellence for Risk Assessment Center (or TERA Center). TERA was organized in 1995 as a nonprofit with a mission to support the protection of public health through the best use of toxicity data. Now as a Center with the Department of Environmental Health at the University of Cincinnati's College of Medicine, we continue to accomplish this mission through independent evaluation of toxicity data and by interpreting and communicating risk assessment information through assessments and websites, organizing peer reviews and consultations, improving risk methods through research, and educating risk managers, assessors, and the public on risk assessment issues. TERA has a strong history of enhancing the use of chemical-specific data to increase the rigor and transparency of evaluations aimed at the prevention of potential human health risks. The TERA Center will maintain this rigor and transparency, but will also mesh its work with the research findings of UC investigators in order to develop the next generation of risk assessment methods based on Toxicology 21 principles. SUPPORTING MISSIONS TERA Center: To support the protection of public health by developing, reviewing and communicating risk assessment values and analyses; improving risk methods through research; and educating risk assessors, managers, and the public on risk assessment issues. UC-DEH: To improve the quality of life by identifying the mechanisms of disease and injury due to environmental exposures and genetic factors, and by developing effective methods of preventions and interventions. 5 2015 TERA Annual Report	Kazakh TERA Цинциннати университетіне қосылды Тәуекелді бағалау үшін Токсикологиядағы Үздік Тәжірибе (TERA) 2015 жылғы 6 шілдеде Цинциннати университетінің (UC) Медицина колледжінің Қоршаған орта денсаулығы бөліміне қосылды. Енді TERA Тәуекелді Бағалау бойынша Токсикология Орталығы (TERA Center) деп аталады. Uzbek TERA Sinsinnati Universitetiga Qo'shildi Riskni baholash uchun Toksikologiya Excellence (TERA) 2015-yil 6-iyul kuni Sinsinnati Universiteti (UC) Tibbiyot Kollejining Atrof-muhit Sog'lig'i bo'limiga qo'shildi. Endi TERA Toksikologiyada Riskni Baholash Markazi (TERA Center) nomi bilan ataladi.

Source: <https://www.industrydocuments.ucsf.edu/docs/mcn0226>

Table 5. Example translations from English to Kazakh and Uzbek on the PATIMT-Bench-Multilingual dataset. The source document contains diverse content types including technical information, structured formatting, and mixed content. TransMLLM successfully translates the content while preserving document structure and technical terminology across multiple low-resource languages with different writing systems (Cyrillic for Kazakh, Latin for Uzbek).